2 A circular disc of radius 11.8 cm is spinning about its centre O in a vertical plane at a rate of 100 revolutions per minute. A plasticine of mass 3.8 g is stuck to the edge of the disc at point P and the line OP is 45 ° from the vertical at the instant, as shown in Fig. 2.1.

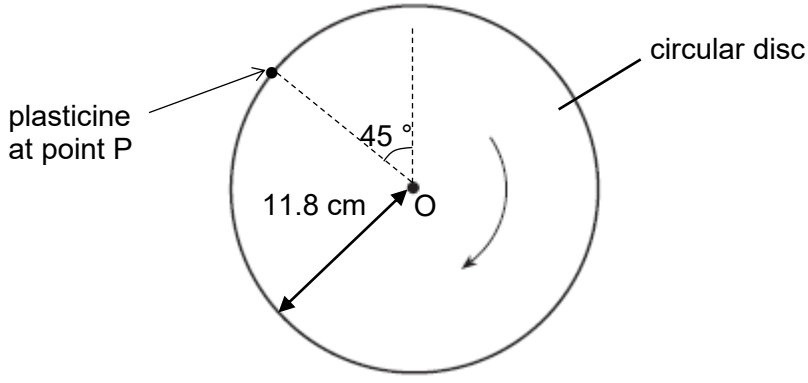

Fig. 2.1

(a) Show that the centripetal force acting on the plasticine is 0.049 N.

[3]

(b) On Fig. 2.2, the weight of the plasticine at P has been drawn. At this instant, the magnitude of the contact force by the disc on the plasticine is equal to the weight of the plasticine. Draw an arrow on Fig. 2.2 to show the contact force by the disc on the plasticine at P. Label this arrow C. [1]

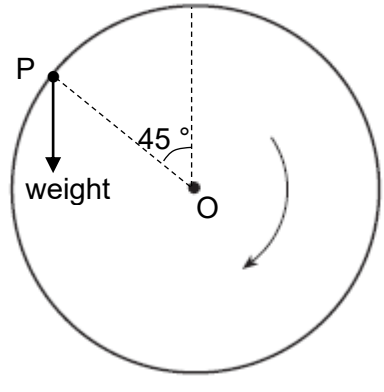

Fig. 2.2

(c) The angular velocity of the disc is increased gradually.

The maximum value of the contact force between the disc and plasticine is 0.23 N.

(i) Explain why the plasticine is most likely to first lose contact with the disc at the lowest point of the revolution.

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…….....

…… [3]

(ii) Hence, determine the angular velocity when the plasticine first loses contact with the disc.

angular velocity = …….. rad s^{-1} [2]